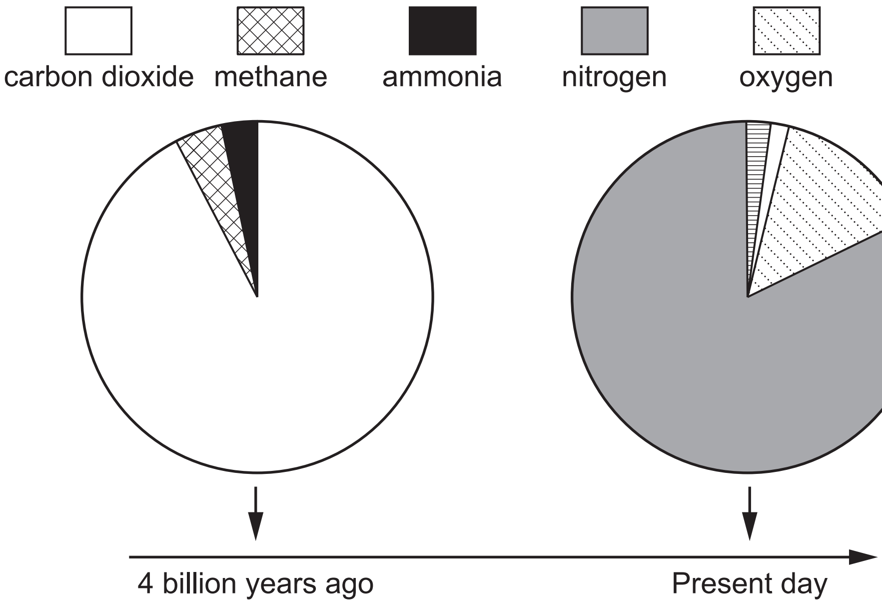
Examiner only

Answer **all** questions.

1. The pie charts show the composition of the air on the Earth 4 billion years ago and the present day.



(a) (i) Give **two** reasons why the amount of carbon dioxide has changed so much over this time period. [2]

1. ....

2. ....

(ii) State **three** other changes shown by the pie charts. [3]

1. ....

2. ....

3. ....

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(b) The table below shows some of the other gases in the air during the present day.

Gas	Chemical formula	Approximate percentage composition (%)
argon	Ar	1.0
hydrogen	H_2	0.00005
methane	CH_4	0.0002
neon	Ne	0.002
nitrous oxide	N_2O	0.00003
ozone	O_3	0.000004

Tick (✓) **three** correct statements about the gases in the air. [3]

- Nitrous oxide contains carbon atoms ☐
- Ozone only contains oxygen atoms ☐
- Hydrogen is an element that occurs as a molecule ☐
- There is a higher percentage of nitrous oxide than methane ☐
- The nitrous oxide molecule contains two oxygen atoms ☐
- Methane is a hydrocarbon ☐

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